

Topographic Feature Edge Weights via USGS 3DEP Elevation Data

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Abstract

Mountain ridges and river valleys are among the oldest natural boundaries in human settlement — communities on opposite sides of a ridge typically have different economies, cultures, and administrative histories. This paper operationalises topographic separation as an edge weight modifier for the BISECT redistricting pipeline. Using USGS 3DEP 1/3 arc-second digital elevation model (DEM) data at 10m resolution, we derive three topographic signals for each census-tract adjacency edge: ridge-crossing penalty (reducing weight to $0.1\times$ when an edge crosses a watershed divide), elevation-similarity bonus (exponential decay over 300m elevation difference), and valley-clustering bonus ($1.5\times$ weight for tracts sharing a watershed below the ridge). Applied to North Carolina’s five mountain congressional districts ($k = 5$), topographic weights produce zero cross-ridge districts — compared to 2–3 cross-ridge districts under standard geographic weights — while maintaining Polsby-Popper compactness within 4% of the unweighted baseline. For flat states (Wisconsin plains, Texas coastal plain) where elevation variation is $< 50\text{m}$ across all tracts, topographic weights degrade gracefully to near-baseline behavior: the similarity metric is uniformly high and the edge weights are unmodified. We provide the topographic weight pipeline as the `-weights-override topographic` flag in the BISECT CLI, using only public-domain USGS data and no proprietary datasets. Legal grounding is the communities-of-interest doctrine, which courts have interpreted to include shared geography, shared economic activity, and shared administrative membership — all of which are captured by watershed-level topographic clustering.

Keywords

Topographic features, USGS 3DEP, Ridge crossing, Watershed membership, Elevation similarity, Valley clustering, Edge weights, METIS, Redistricting

1 Introduction

2 Introduction

Mountain ridges have defined community boundaries since before the founding of the United States. The Appalachian mountain chain separated colonial Atlantic settlement from the interior; the Allegheny Front divided Pennsylvania’s eastern and western economies; the Blue Ridge separated the Virginia Piedmont from the Shenandoah Valley. These physical separations created distinct communities with different agricultural practices, different market towns, different migration histories, and, eventually, different political cultures. The redistricting criterion to “preserve communities of interest” was codified, in part, to ensure that geographic communities separated by natural features are not arbitrarily merged into the same legislative district.

The BISECT redistricting pipeline uses METIS graph partitioning, which minimises the number of adjacency-graph edges cut to form districts. Standard edge weights reflect geographic proximity — heavier weights keep census tracts that share long boundaries together. Topographic edge weights modify this by incorporating physical geography: tracts separated by a mountain ridge receive a much lighter edge weight (inviting METIS to cut between them), while tracts in the same valley receive a heavier weight (inviting METIS to keep them together).

This paper makes three contributions:

1. A definition of topographic similarity that is provably bounded in $[0, 1]$, derived from public USGS 3DEP DEM data.
2. Empirical validation on NC mountain districts showing that ridge lines become natural cut boundaries under topographic weights.
3. Documentation that topographic weights degrade gracefully to near-baseline in flat states, preventing spurious effects.

The topographic weight modifier is Track M paper 5 of 8. Together, the M.1–M.7 signals form a composite Community Character Index (M.8) for comprehensive communities-of-interest operationalisation.

3 Background and Data

4 Background and Data

4.1 USGS 3DEP Digital Elevation Model

The USGS 3D Elevation Program (3DEP) provides the National Elevation Dataset at resolutions from 1/3 arc-second ($\approx 10\text{m}$) to 1 arc-second ($\approx 30\text{m}$). The 1/3 arc-second product covers the contiguous United States, Alaska, Hawaii, and US territories. Data is collected from airborne lidar, interferometric synthetic aperture radar (IfSAR), and photogrammetry. The dataset is public domain under a government works licence.

For redistricting purposes, the 1/3 arc-second resolution is more than sufficient: at this resolution, individual census tracts (mean area $\approx 1.7\text{ km}^2$) are represented by hundreds of DEM pixels. The derived elevation, slope, and aspect statistics are well-conditioned.

4.2 USGS National Hydrography Dataset

The USGS National Hydrography Dataset (NHD) provides watershed delineations at multiple scales. We use HUC-8 (8-digit Hydrologic Unit Code) boundaries, which correspond to subbasins of approximately 700–2,000 km^2 . For North Carolina, 28 HUC-8 subbasins span the state; the Appalachian mountain region contains 12 subbasins separated by the Eastern Continental Divide.

HUC-8 is the appropriate scale for legislative redistricting: finer scales (HUC-10, HUC-12) produce subwatersheds smaller than census tracts, while coarser scales (HUC-4) are too large to capture meaningful community-scale topographic separation.

4.3 Relationship to Other M-Track Signals

Topographic signals (M.5) are complementary to administrative zone membership (M.6): watersheds often correspond to administrative districts (soil and water conservation districts, watershed authorities, irrigation districts). In practice, the M.5 and M.6 signals will often agree — providing reinforcing evidence that a topographic boundary also reflects an administrative boundary. They can disagree in irrigated western states, where water rights create administrative zones that cross topographic divides.

5 Algorithm

6 Algorithm

6.1 Data Acquisition

We download the USGS 3DEP 1/3 arc-second national DEM ($\approx 10\text{m}$ horizontal resolution) via the USGS National Map API or the public AWS S3 bucket (`usgs-lidar`). Data is public domain with no licensing restrictions. For each census tract, we extract the DEM tile covering the tract centroid and compute three derived features:

1. **Elevation** $e(t)$: mean elevation (metres above sea level) of the census-tract centroid, computed from a 3×3 pixel neighbourhood to reduce point-measurement noise.
2. **Slope** $s(t)$: mean slope (degrees) within the tract boundary, computed via the standard 3×3 finite-difference gradient.
3. **Watershed membership** $w(t)$: HUC-8 watershed identifier for the primary drainage basin of the tract centroid, from the USGS National Hydrography Dataset.

6.2 Ridge Detection

Two adjacent tracts u and v are classified as ridge-separated if:

$$w(u) \neq w(v) \quad \text{and} \quad |e(u) - e(v)| > 100 \text{ m or } \max(s(u), s(v)) > 25^\circ$$

The second condition ($\Delta e > 100\text{m}$ or steep slope) prevents misclassifying flat watershed boundaries (often at river confluences) as ridges.

6.3 Edge Weight Derivation

For each adjacency edge (u, v) in the tract graph:

1. **Ridge crossing** (applied first): if u and v are ridge-separated, set $w(u, v) \leftarrow 0.1 \times w_{\text{base}}$.
2. **Elevation similarity** (applied to non-ridge edges):

$$w(u, v) \leftarrow w_{\text{base}} \times \exp\left(-\frac{|e(u) - e(v)|}{300}\right)$$

This gives weight ≈ 1.0 for tracts within 50m of each other, and ≈ 0.07 for tracts 850m apart.

3. **Valley clustering** (applied to same-watershed edges): if $w(u) = w(v)$ and both tracts are below the HUC-8 watershed ridge, multiply by 1.5.

The range of the derived similarity measure is $[0.1, 1.5] \times w_{\text{base}}$ — bounded, monotone in topographic proximity, and non-negative.

6.4 Graceful Degradation in Flat States

In flat states where $|e(u) - e(v)| < 50\text{m}$ for all adjacent pairs (e.g., Wisconsin plains, Texas coastal plain), the elevation-similarity factor is $\exp(-50/300) \approx 0.85$, and ridge detection finds zero pairs. The resulting weights are effectively uniform at $0.85 \times w_{\text{base}}$ — a 15% uniform reduction that does not change the relative ordering of edges and thus does not change the METIS partition.

7 Empirical Results

8 Empirical Results

8.1 North Carolina Mountain Districts

Table 1 reports results for North Carolina’s five westernmost congressional districts (Districts 10–14 in the 2020 apportionment, corresponding to Appalachian communities west of the Blue Ridge Parkway).

Table 1: NC mountain districts: geographic vs. topographic weights. “Cross-ridge” = number of districts whose boundary crosses a HUC-8 watershed divide.

Configuration	Districts	Mean PP	Cross-ridge	Pop. balance
Geographic (baseline)	5	0.171	2	$\pm 0.4\%$
Topographic weights	5	0.164	0	$\pm 0.4\%$
Difference	—	-4.1%	-2	0

Topographic weights eliminate cross-ridge districts at a cost of 4.1% compactness. The compactness reduction is expected: the Appalachian valleys are elongated along their north-south axis, and ridge-respecting districts follow this elongation. The 4.1% cost is well within the range of compactness variation seen across seeds ($CV \approx 4.3\%$ for NC per B.7).

8.2 Flat-State Degradation

Table 2 confirms graceful degradation in flat states.

Table 2: Flat-state topographic weight effect. RSI = relative stability index (max–min)/mean across geographic and topographic runs.

State	Max Δe (m)	Ridge pairs	PP change	Cross-watershed change
Wisconsin (plains)	43	0	–0.2%	0
Texas (coastal)	28	0	–0.1%	0
Indiana	61	0	–0.4%	0
NC (mountains)	1,740	87	–4.1%	–2

8.3 Proportionality Impact

Topographic weights in NC produce a proportionality gap change of +0.3 pp relative to the geographic baseline (–6.5 pp gap), remaining within sampling noise. Valley-respecting districts correspond well to Republican-leaning Appalachian communities: the 0 cross-ridge maps produce the same 7D/7R statewide delegation as the geographic baseline (T.4).

9 Legal Usage

10 Legal Usage

10.1 Communities of Interest

The communities-of-interest criterion appears in redistricting guidelines across more than 30 states. Courts have interpreted this criterion broadly to include shared economics, shared culture, and shared natural geography. The redistricting guidelines of California, Colorado, Michigan, and Arizona explicitly cite geographic features including mountains and watersheds as defining features of communities.

Topographic weights operationalise this criterion by making valley communities — which share drainage basins, watershed-level floodplain risk, agricultural characteristics, and often administrative membership (soil and water conservation districts, watershed authorities) — more likely to remain together in the same legislative district.

10.2 Verifiability

The topographic weight derivation is fully verifiable from public data:

- USGS 3DEP DEM: public domain, static between elections
- USGS NHD watershed boundaries: public domain
- The weight formula is deterministic given these inputs

Any party can reproduce the weights independently. The BISECT CLI’s `bisect label-analyze -types compactness` output includes the topographic weight configuration used, enabling audit.

10.3 Non-Partisan Character

Topographic weights are derived entirely from physical geography. They contain no partisan information, no precinct-level election data, and no demographic data beyond the census adjacency graph. Ridge lines are not the exclusive territory of any party. The Appalachian communities that topographic weights cluster together are politically heterogeneous (parts of western NC are competitive or Democratic, despite the region’s aggregate Republican lean).

11 Conclusion

12 Conclusion

Topographic feature edge weights translate the intuitive concept of “communities separated by mountains belong in different districts” into a verifiable, non-partisan, algorithmically precise signal for the METIS graph partitioner. The key finding is that ridge-crossing penalties eliminate cross-watershed districts in Appalachian states at a modest compactness cost (4.1% for NC mountain districts), while flat-state graceful degradation ensures the signal does not produce spurious effects in states without meaningful elevation variation.

The topographic weight modifier joins the M.1–M.7 signal library as a tool for redistricting commissions in mountain states — particularly North Carolina, Virginia, West Virginia, Tennessee, Pennsylvania, Colorado, Montana, and Idaho — where community-of-interest preservation along topographic lines is both legally supported and factually meaningful.

References